



Data Science and AI Lab

May 2026 Term

Milestone 1 Report

Group 4

Description of problem and expected users

Problem Statement

Job search is fragmented because candidates use separate tools for job discovery, resume tailoring, ATS checking, and cover letters.

This makes the process time-consuming, repetitive, and difficult to personalize for each role.

Proposed Solution

The system will use a multi-agent AI pipeline to convert a resume into a structured candidate profile.

It will recommend suitable roles, rank matching jobs, and generate tailored resumes and cover letters.

Stakeholders:

Primary: job seekers, fresh graduates, career changers, working professionals.

Secondary: recruiters and employers.



Problem Scope

Scope

The project includes resume parsing, where the uploaded resume is converted into structured information such as skills, education, experience, projects, and job titles.

It also includes candidate profile creation, so all agents can use one common profile instead of taking repeated input from the user.

The system will recommend suitable career roles, discover relevant job listings, rank jobs based on profile fit, and provide ATS-style scoring.

It will also generate tailored resumes and personalized cover letters for selected job roles through a web-based interface.

Out of scope

The project will not verify whether a job post or company is genuine, because employer background verification is outside the current system goal.

It will not extract private contact details of recruiters or directly message recruiters

The system will not create fake skills, false experience, or unsupported achievements in resumes or cover letters.

It will also not include interview preparation and fully automatic job applications without user review.



Why this problem matter and existing solutions

Importance and cause of the problem

Why It Matters:

Job search is becoming more competitive, time-consuming, and AI-driven.

Many candidates are actively looking for jobs, but finding the right role is becoming harder.

Where It Arises:

This problem appears on online job platforms such as LinkedIn, Naukri, Indeed, Glassdoor, company career pages, and ATS-checking tools.

Even with many platforms available, candidates still struggle to identify which jobs truly match their profile.

Key Challenges:

- Too many job listings create confusion and overload.
- Candidates must repeatedly tailor resumes and cover letters for each role.
- Most candidates do not get clear feedback on how well their resume matches a job description.

Project Importance:

This project helps candidates move from simply finding jobs to finding better-fit jobs.

It supports them in preparing stronger, more personalized application material for each role.



Current solution available

How It Is Addressed Today

The current job search process is handled by separate tools for different tasks.

Existing solution

- Job boards like LinkedIn, Naukri, Indeed, and company career pages help users find openings.
- Resume builders and LLM tools help write resumes and cover letters.
- ATS tools like Jobscan and SkillSyncer compare resumes with job descriptions.
- Career guidance tools give general role suggestions.

Main limitation

These tools work separately and are not connected through one candidate profile. Because of this, users still have to manually compare jobs, tailor resumes, check ATS scores, and prepare applications using different platforms.



Metrics & Success Definition

Quantitative Metrics

- **Precision, Recall, F1-Score:** Used to check resume extraction accuracy.
- **Top-K Accuracy and MRR:** Used to evaluate career role recommendations.

Qualitative Metrics:

- **LLM-as-Judge:** Rates resume tailoring, cover letters, and recommendations on relevance, clarity, personalization, and factual accuracy.
- **Human Review:** Checks whether generated content is professional, specific, and based on the candidate's real profile.

Success Definition:

- Success means accurate resume parsing, useful career recommendations, and high-quality tailored resumes and cover letters.
- The system should improve application quality without adding fake or unsupported information.



Shortfall in Current Solutions

Current shortfall

Today's job search tools are disconnected. Candidates must use separate platforms for job search, resume writing, ATS checking, and cover letters.

Key shortfall:

- Job discovery is mostly limited to crowded job boards.
- Candidates manually copy information between different tools.
- AI-generated resumes and cover letters are often generic.
- There is limited feedback on how well a resume matches a job.
- Career guidance is usually general and does not update with the candidate's growth.

Opportunity:

Our project connects all these steps through one shared candidate profile. This allows a multi-agent AI system to support the user from job discovery to tailored application material in one workflow.



Project Contribution

Main Contribution

This project is not focused on beating academic benchmarks.
It focuses on building a practical and usable AI-based job search workflow.

Contribution Areas:

- Combines resume parsing, job matching, ATS scoring, resume tailoring, and cover letter generation in one pipeline.
- Uses grounded generation, so outputs are based only on the candidate profile and job description.
- Focuses on quality applications instead of blind auto-apply.



Proposed System Architecture / Workflow

How our structure look like

The system uses a **multi-agent AI pipeline** built around one **shared candidate profile**.

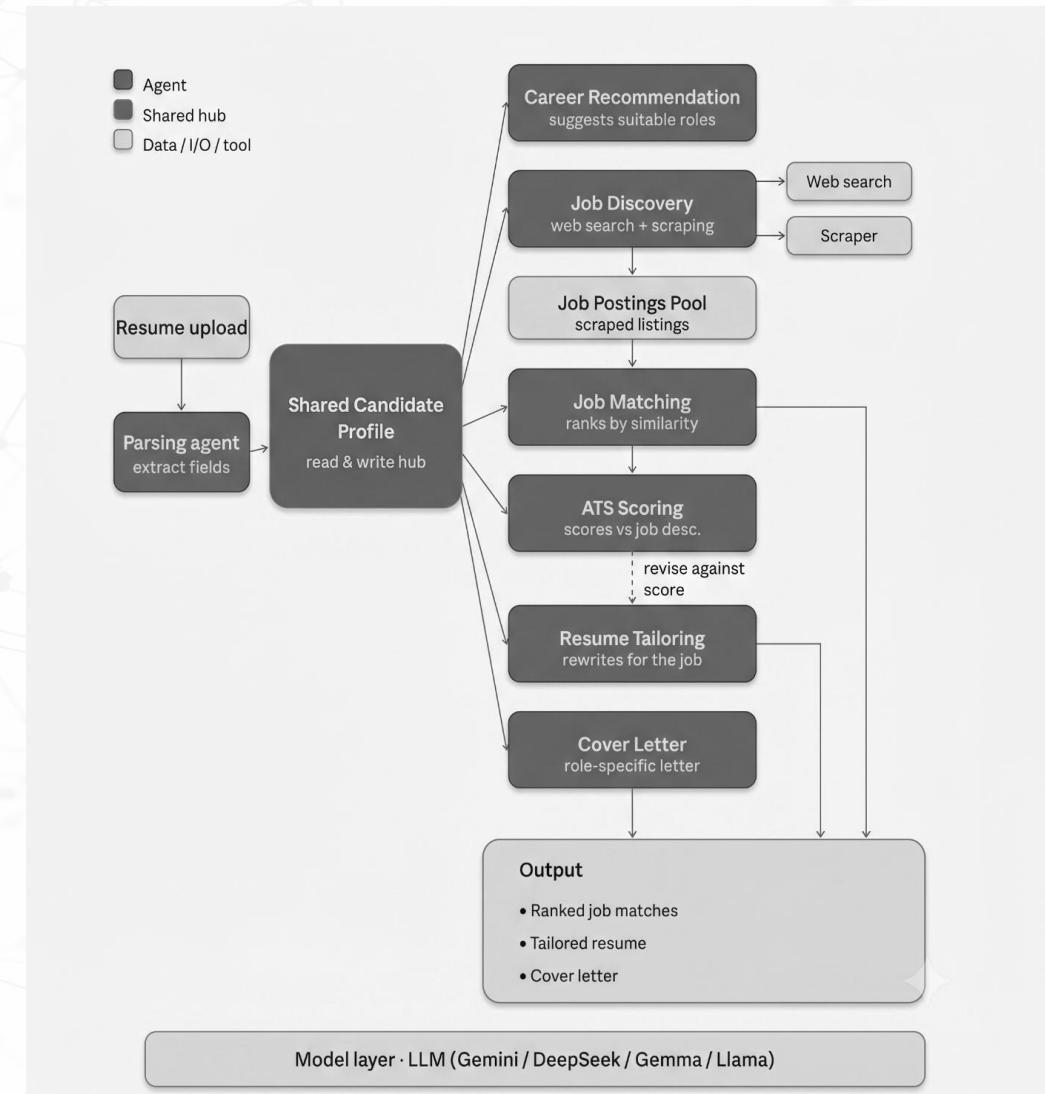
The user uploads a resume, and the **Parsing Agent** extracts skills, education, experience, projects, and job titles into a structured profile. The **Career Recommendation Agent** suggests suitable roles from this profile.

The **Job Discovery Agent** searches public job sources and creates a pool of job postings.

The **Job Matching Agent** ranks jobs by similarity with the candidate profile.

The **ATS Scoring Agent** checks how well the resume matches a selected job description.

Finally, the **Resume Tailoring Agent** and **Cover Letter Agent** generate role-specific application material.



Preliminary Dataset Plan and Challenges

Dataset

Pipeline Component	Expected Data Source	Purpose
Resume parsing	Public Kaggle resume datasets, open-source resume samples.	To extract skills, education, experience, projects, certifications, and job titles
Candidate profile creation	Parsed resume data	To create a structured candidate profile used by all agents
Job discovery and matching	Public job description datasets, company career pages etc.	To collect job title, company, location, required skills, experience, job type, and description
Career Recommendation	O*NET, ESCO, and public skill-role taxonomies	To map candidate skills and experience to suitable job roles
Evaluation data	Small, curated set of resumes, job descriptions, expected role suggestions etc.	To evaluate output quality and system usefulness

Challenges

- Data availability
- Privacy
- Dynamic job postings



Evidence of Working Approach

Purpose at This Stage:

The aim is not large-scale testing, but to show that the Generative AI pipeline works clearly on sample resumes and job descriptions

Evidence of success:

- Resume details like skills, education, experience, projects, and job titles are extracted correctly.
- Extracted information is converted into a structured candidate profile.
- The system suggests suitable and explainable career roles.
- Relevant jobs are ranked higher based on candidate profile fit.
- Tailored resumes show better ATS-style alignment with the job description.
- Generated resumes and cover letters are personalized and do not include fake information.
- The complete flow works through the web interface from resume upload to application material generation.

Success Definition

The project will be successful if it reduces manual effort and produces better, more personalized job application material than using separate tools manually.



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